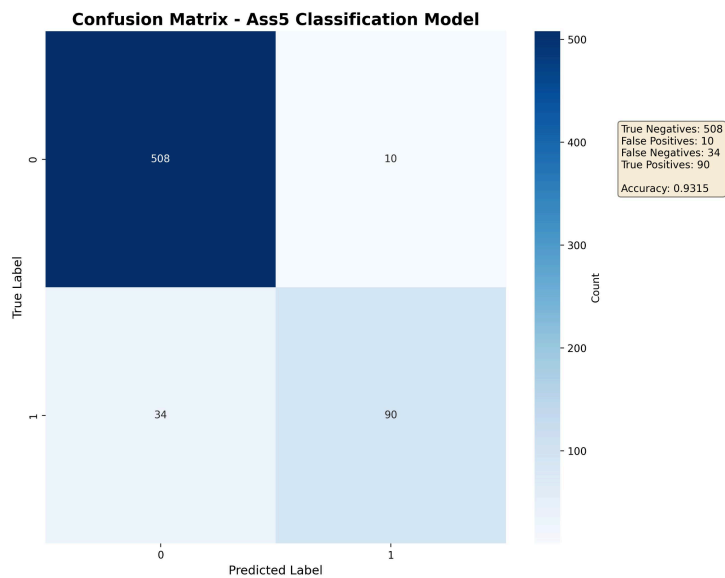


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Results:

Data Preparation

I started with the full dataset in `ass5data.csv` and split it into training and test sets using `make_split.py`. The split used `test_size=0.2` and `random_state=42`, with stratification on `ProdTaken` so both sets kept the same class balance. This produced 2,566 rows for training and 642 rows for testing. In both sets, class 0 is about 80.7% and class 1 is about 19.3%. After splitting, I cleaned the data by trimming text spaces, fixing inconsistent category names, removing rows with missing target values, and filling missing feature values (median for numeric columns, most frequent value for categorical columns).

Analysis

The dataset combines customer profile details and customer interaction details. Profile features include age, occupation, city tier, and income. Interaction features include pitch duration, number of follow-ups, and satisfaction score. These interaction features are likely strong predictors of purchase behavior, while profile features help add context. Since the dataset has more non-buyers than buyers, I used more than accuracy alone to judge performance, especially recall and AUC.

Feature Extraction

The target variable is `ProdTaken` (0 = not purchased, 1 = purchased). All other columns were used as input features. Categorical columns were converted using one-hot encoding so the model could learn category-specific patterns. Numeric columns were kept numeric and imputed when missing. This preprocessing was built into a pipeline so the same feature logic was applied consistently during both training and testing.

Building Model

I trained three tree-based classifiers: Decision Tree, Random Forest, and Extra Trees. I used class weighting to help with class imbalance, and for Random Forest and Extra Trees I used 800 trees with `max_features='sqrt'`. Inside the training set, I used an additional 80/20 split for

validation and tuned a probability threshold from 0.05 to 0.95. The best model was Extra Trees, and the selected threshold was 0.455. I then retrained this final model on the full training set and saved it for inference.

Evaluation Results

On validation data, the model achieved accuracy 0.9086, recall 0.6364, precision 0.8514, F1-score 0.7283, and AUC 0.9266. **On the final test set, the model achieved accuracy 0.9315 (about 93%),** recall 0.7258, precision 0.9000, F1-score 0.8036, and AUC 0.9535. The confusion matrix on test data was: True Negatives = 508, False Positives = 10, False Negatives = 34, and True Positives = 90. These results show strong overall accuracy and strong precision, with moderate recall for the positive class.